FLEXOPTIX

Comprehensión y optimización de la eficiencia de transceptores ópticos

utilizando métricas internas para mejorar el ahorro de energía

Dr. Gerhard Stein y Thomas Weible

Contenido

VDM → Láser → TEC → Power Budget → Eficiencia → Resultados

- **Versatile Diagnostics Monitoring**
- **Láser**: Discusión sobre el rendimiento de lásers (trc)
- **TEC:** El refrigerador termoeléctrico
- **Power Budget:** Como la FLEXBOX administra la energía de transceptores
- **Eficiencia:** Una definición
- **Resultados**

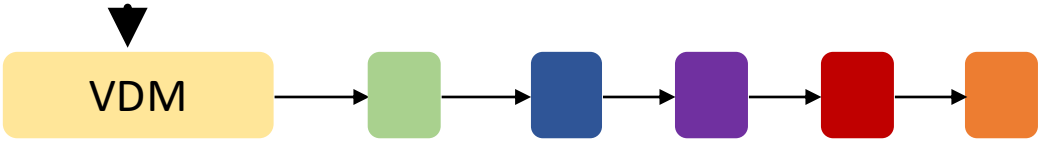
Devices under Test (DUT)

Transfer/Receiver -> Transceiver -> Trc:

- 100G QSFP28 (500 m)
- 400G QSFP-DD Coherent ZR (120 km)
- 800G QSFP-DD (2 km)
- 800G QSFP-DD (500 m)

Switches:

- Cisco **93600CD-GX** NX OS 10.5.3 (F)
- Cisco **C9500** IOS XE 17.14.1
- Juniper **QFX5120** JunOS 25.R2R1.9



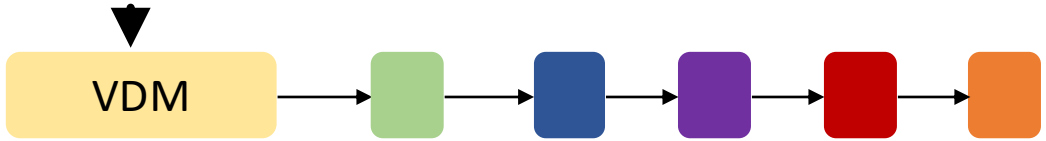
Versatile Diagnostic Monitoring

Type ID	Observable Type	Instance Type	Data Type	Unit Scale	Unit
0	Not Used indicator[1]	N/A	N/A		
1	Laser Age (0% at BOL, 100% EOL) (Media Lane)	Basic	U16	1	%
2	TEC Current (Module)	Basic	S16	100/32767	%
3	Laser Frequency Error (Media Lane)	Basic	S16	10	MHz
4	Laser Temperature (Media Lane)	Basic	S16	1/256	C
5	SNR (dB) Media Input (Media Lane) –- see section 7.1.4	Basic	U16	1/256	dB
6	SNR (dB) Host Input (Lane) –- see section 7.1.4	Basic	U16	1/256	dB
7	PAM4 Level Transition Parameter Media Input (Media Lane)	Basic	U16	1/256	dB
8	PAM4 Level Transition Parameter Host Input (Lane)	Basic	U16	1/256	dB
9	Pre-FEC BER Minimum Sample Media Input (Data Path)	Statistic	F16	N/A	
10	Pre-FEC BER Minimum Sample Host Input (Data Path)	Statistic	F16	N/A	
11	Pre-FEC BER Maximum Sample Media Input (Data Path)	Statistic	F16	N/A	

<u>En comparación a Digital Diagnostics Monitoring (DDM)</u>

Byte	Bit	Name	Description
22	All	Temperature MSB	Internally measured temperature (MSB)
23	All	Temperature LSB	Internally measured temperature (LSB)
24-25	All	Reserved	
26	All	Supply Voltage MSB	Internally measured supply voltage (MSB)
27	All	Supply Voltage LSB	Internally measured supply voltage (LSB)

Sources: [2, 15]



Versatile Diagnostic Monitoring

DDM

Leer datos

Byte	Bit	Name
22	All	Temperature MSB
23	All	Temperature LSB

Calcular valor

VDM

Para cada uno de los desc.

Page	Subject Area
20h	Descriptors for VDM Instances 1-64 (Group 1)
21h	Descriptors for VDM Instances 65-128 (Group 2)

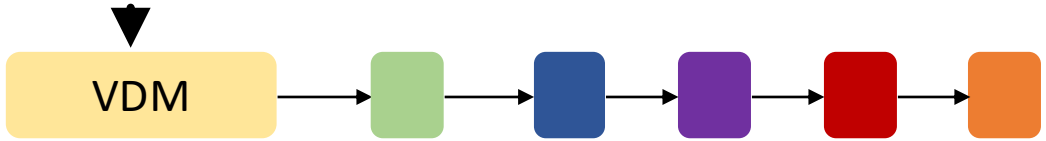
¿Cuál es?

Type ID	Observable Type
0	Not Used indicator[1]
1	Laser Age (0% at BOL, 100% EOL) (Media Lane)
2	TEC Current (Module)
3	Laser Frequency Error (Media Lane)
4	Laser Temperature (Media Lane)
5	SNR (dB) Media Input (Media Lane) –- see section 7.1.4

¿Dónde está el valor?

Page	Byte	Register Name
24h	128-129	VDMSample1
	130-131	VDMSample2

Calcular valor



VDM – ¿Suporte?

Sólo DDM (valores actuales):

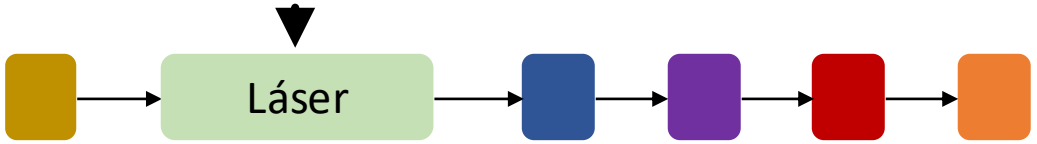
```
{master:0}
root@QFX5120-48Y> show interfaces  diagnostics optics et-0/0/49
Physical interface: et-0/0/49
    Module temperature                          :  37 degrees C / 98 degrees F
    Module voltage                              :  3.2700 V
    Module temperature high alarm               :  Off
```

Juniper QFX5120
JunOS 25.2R1.9
No hay suporte ☹

Sólo DDM (límites):

```
                                MODULE THRESHOLDS
                        high alarm      high warning    low warning     low alarm
Temperature     C           +076.000        +073.000        -03.000       -06.000
Voltage         V           003.6000        003.5500        003.0500      002.9500

                                LANES THRESHOLDS
                        high alarm      high warning    low warning     low alarm
Bias Current    mA          130.0000        125.0000        015.0000      010.0000
Transmit power  mW          003.1623        002.5119        000.5129      000.4074
Receive power   mW          004.4668        003.5481        000.1819      000.1288
```

Cisco C9500 **IOS XE 17.14.1**
No hay suporte ☹



VDM – ¿Suporte?

Cisco Nexus **93600CD-GX** NX OS 10.5.3 (F) 400G Coherent ZR

```
--------------------------------------------------------------------------------
                    Current                    Alarms                  Warnings
                    Measurement             High        Low         High        Low
--------------------------------------------------------------------------------
Temperature           48.89 C             78.00 C     -8.00 C     73.00 C     -3.00 C
Voltage                3.30 V              3.63 V      2.97 V      3.46 V      3.13 V
Current                 N/A                N/A         N/A         N/A         N/A
Tx Power                N/A                N/A         N/A         N/A         N/A
                                           N/A         N/A         N/A         N/A
Laser temperature      49.98 °C            N/A         N/A         N/A         N/A
                                           N/A         N/A         N/A         N/A
Pre-FEC BER         1.00e+00               N/A         N/A         N/A         N/A
Post-FEC BER        1.00e+00               N/A         N/A         N/A         N/A
CD (Short Link)        0.00 ps/nm          N/A         N/A         N/A         N/A
CD (Long Link)         0.00 ps/nm          N/A         N/A         N/A         N/A
Diff. group delay      0.00 ps             N/A         N/A         N/A         N/A
SOPMD                  0.00 ps^2           N/A         N/A         N/A         N/A
PDL                    0.00 dB             N/A         N/A         N/A         N/A
OSNR                   0.00 dB             N/A         N/A         N/A         N/A
ESNR                   0.00 dB             N/A         N/A         N/A         N/A
Carrier freq off       0.00 MHz            N/A         N/A         N/A         N/A
Err Vector Mag.        0.00 %              N/A         N/A         N/A         N/A
SOP Rate of Chg        0.00 krad/s         N/A         N/A         N/A         N/A
Laser bias           227.62 mA             N/A         N/A         N/A         N/A
SOPMD LO GR            0.00 ps^2           N/A         N/A         N/A         N/A
Modulation Err R     385.30 dB             N/A         N/A         N/A         N/A
Clock recovery         0.00 %              N/A         N/A         N/A         N/A
Transmit Fault Count = 0
--------------------------------------------------------------------------------
Note: ++  high-alarm; +  high-warning; --  low-alarm; -  low-warning
```

CMIS VDM

Type ID	Observable Type
0	Not Used indicator[1]
1	Laser Age (0% at BOL, 100% EOL) (Media Lane)
2	TEC Current (Module)
3	Laser Frequency Error (Media Lane)
4	Laser Temperature (Media Lane)

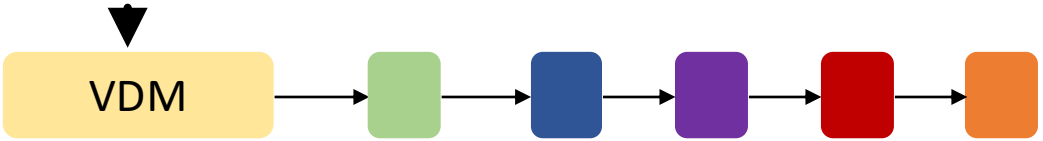
… but …

7.1.6.3 Laser Temperature

If supported, this monitor observes the laser temperature difference between the target laser temperature for a cooled laser, and the actual current temperature.

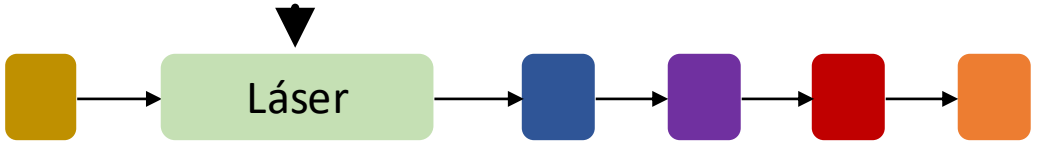
Lo que significa que 50.0 °C – 0.02 °C = 49.98 °C

Valor de **50 °C** es un valor típico para láser [10]; -
→ suposición del switch.



¿VDM suporte? ¿Cómo descubrirlo?

- Dispositivos para la red no siempre lo mencionan directamente
- Checa Laser Temperature y TEC Current
  - Hay más valores, pero éstos también pueden estar en otros registros (BER and OSNR)
  - Por medio de VDM se escoge lo que el transceptor suporta, pero en la mayoría de los casos **Laser temperature** y **TEC current** son los más usados
- Acceso a las páginas de memoria 20h - 2Fh

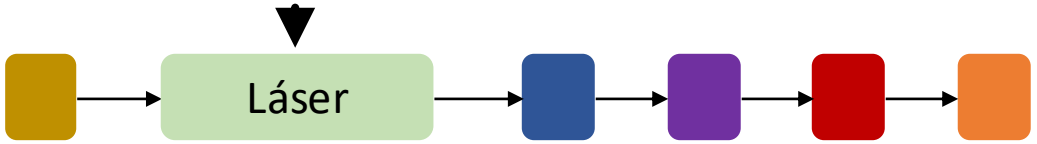


Consumo de energía del láser

- **Ningún** acuerdo o especificación demanda que el voltaje del láser sea reportado → El fabricante tiene esa información.
- $Pow_{cons} = V_{fwd} \times bias$
- Eficiencia electro-óptica = $\frac{Tx_{OptPow}}{Pow_{cons}} < 0.1$
- En nuestro software mapeamos el valor dependiendo del tipo de láser dado

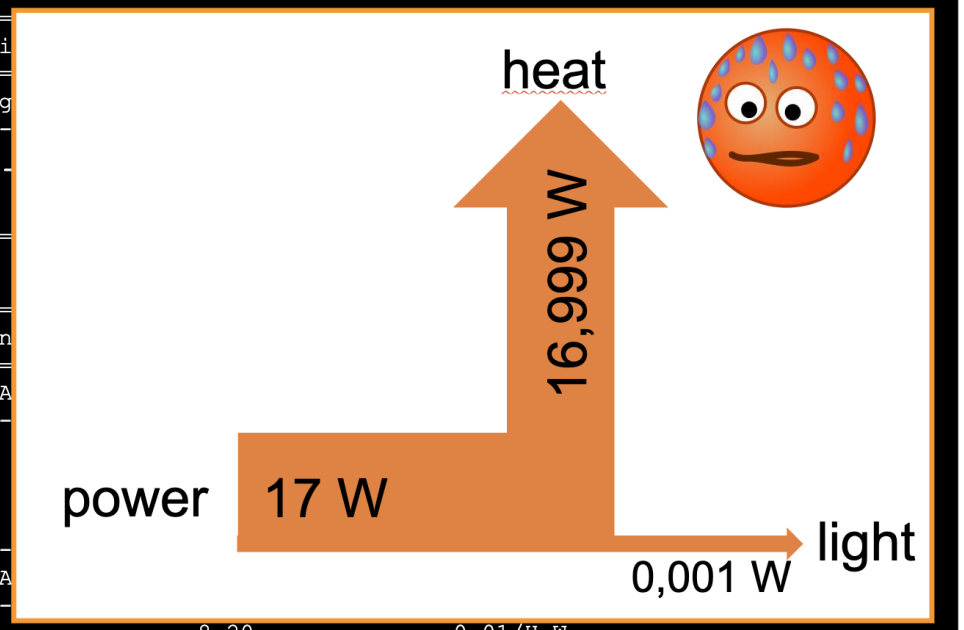
```
laserTypeToVoltage = {
    'EML': 1.3,
    'VCSEL': 1.85,
    'DFB': 1.5
}
```

Source: [16]

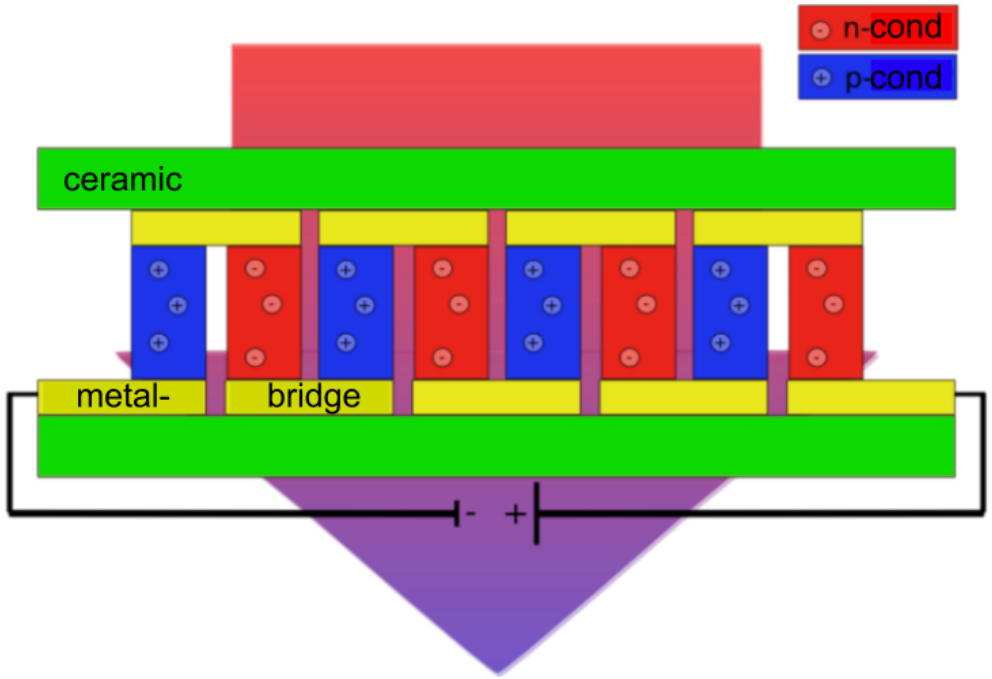


Excursión: Transceptor coherente (2023/2024)

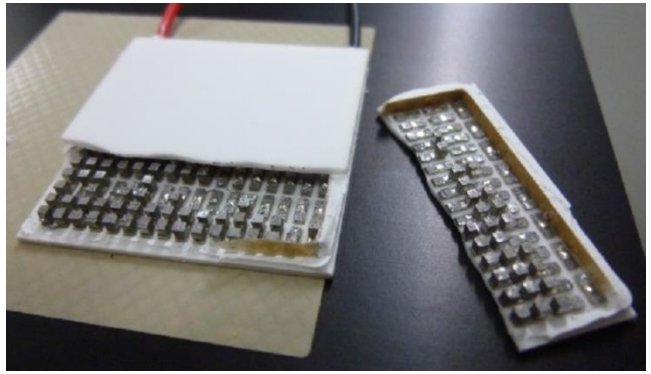
```
still show port 8/1/c7, DDM should be known by now

...

==========================================
Transceiver Digital Diagnostic Monitori
==========================================
                              Value Hig
------------------------------------------
Temperature(C)  +48.0
Supply Voltage (V)              3.26
==========================================


==========================================
Transceiver Lane Digital Diagnostic Mon
==========================================
                                 High A
------------------------------------------
Lane Tx Output Power (dBm)
Lane Rx Optical Pwr (avg dBm)


------------------------------------------
Lane ID Temp(C)/Alm       Tx Bias(mA)/A
------------------------------------------
    1            -                -              -8.20              0.01/H-W
==============================================================================

...
```



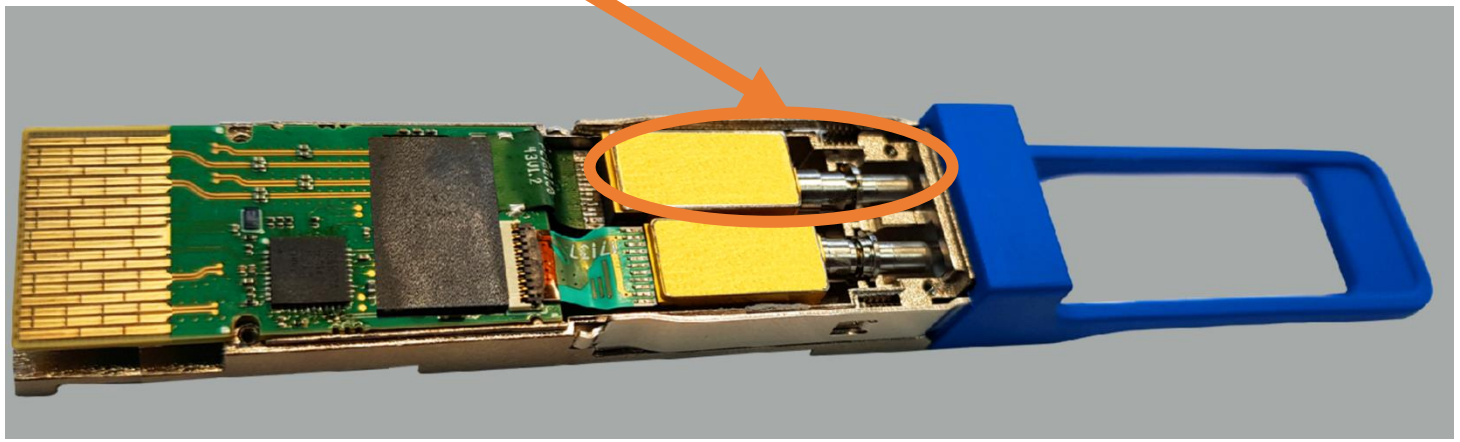
Refri eléctrico (TEC)



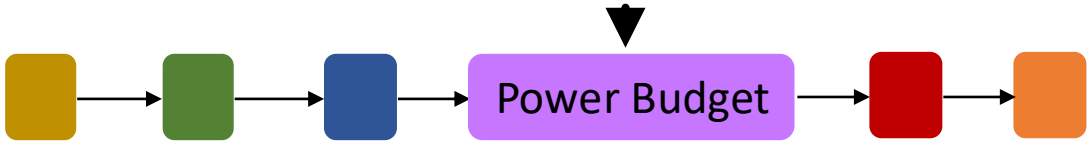
Un elemento Peltier típico en escala grande …



… el TEC es muy pequeño → TOSA



Sources: [10]



Propuesto energético

Bit	Device Type
7-6	00: Power Class 1 (1.5 W max.)
	01: Power Class 2 (2.0 W max.)
	10: Power Class 3 (2.5 W max.)
	11: Power Class 4 (3.5 W max.) and Power Classes 5, 6 or 7
5	Power Class 8 implemented (Max power declared in byte 107)
4	0: No CLEI code present in Page 02h
	1: CLEI code present in Page 02h
3	0: No CDR in Tx, 1: CDR present in Tx
2	0: No CDR in Rx, 1: CDR present in Rx
1-0	00: Power Classes 1 to 4
	01: Power Class 5 (4.0 W max.) See Byte 93 bit 2 to enable.
	10: Power Class 6 (4.5 W max.) See Byte 93 bit 2 to enable.
	11: Power Class 7 (5.0 W max.) See Byte 93 bit 2 to enable.

Byte	Bits	Field Name	Field Description	Type
200	7-5	ModulePowerClass[1]	000: Power class 1 001: Power class 2 010: Power class 3 011: Power class 4 100: Power class 5 101: Power class 6 110: Power class 7 111: Power class 8	RO Rqd.
	4-0	-	**Reserved**	RO
201	7-0	MaxPower	Maximum power consumption in multiples of 0.25 W rounded up to the next whole multiple of 0.25 W	RO Rqd.

Note 1: See relevant hardware specification for maximum power allowed in each Power class

SFF8636 (QSFP+, QSFP28)

↓

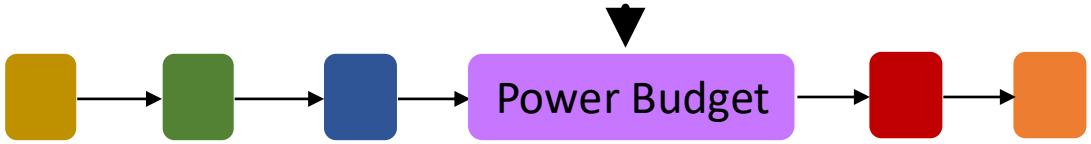
Energía máxima especificada

CMIS (QSFP-DD, OSFP)

↓

El proveedor puede para clase decidir que valor máximo poner

Source: [1,2]



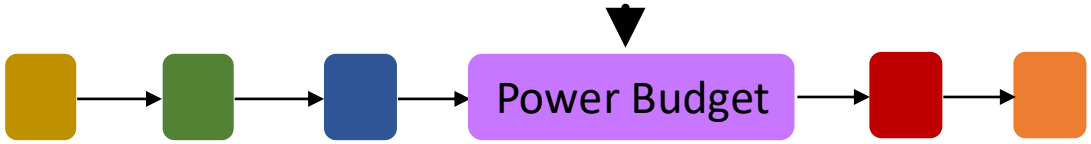
Propuesto energético

Ejemplo: D.CO164HG.2.yTM
(480km @ Multirate 400G SR8 ZR+)

Nuestro **shop** te muestra el consumo máximo

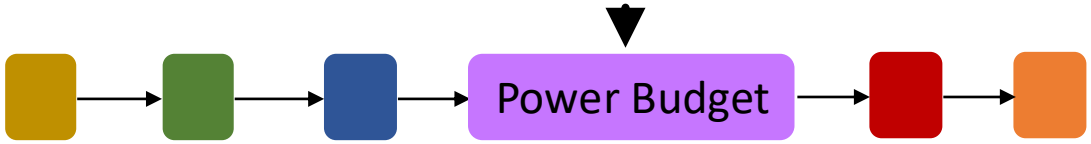
DIGITAL DIAGNOSTIC MONITORING (DDM)	Yes, internally calibrated
POWER CONSUMPTION	22 W
CDR	TX and RX
SGMII	No
INBUILT FEC	Yes, OFEC
POWERBUDGET (DB)	15 dB
TRANSMIT MIN/MAX PER LANE	-5 dBm / 3 dBm
RECEIVER MIN/MAX PER LANE	-12 dBm / 0 dBm (overload) @400G
WAVELENGTH TX (TYPICAL)	tunable Coherent high-power DWDM

Source: [18]

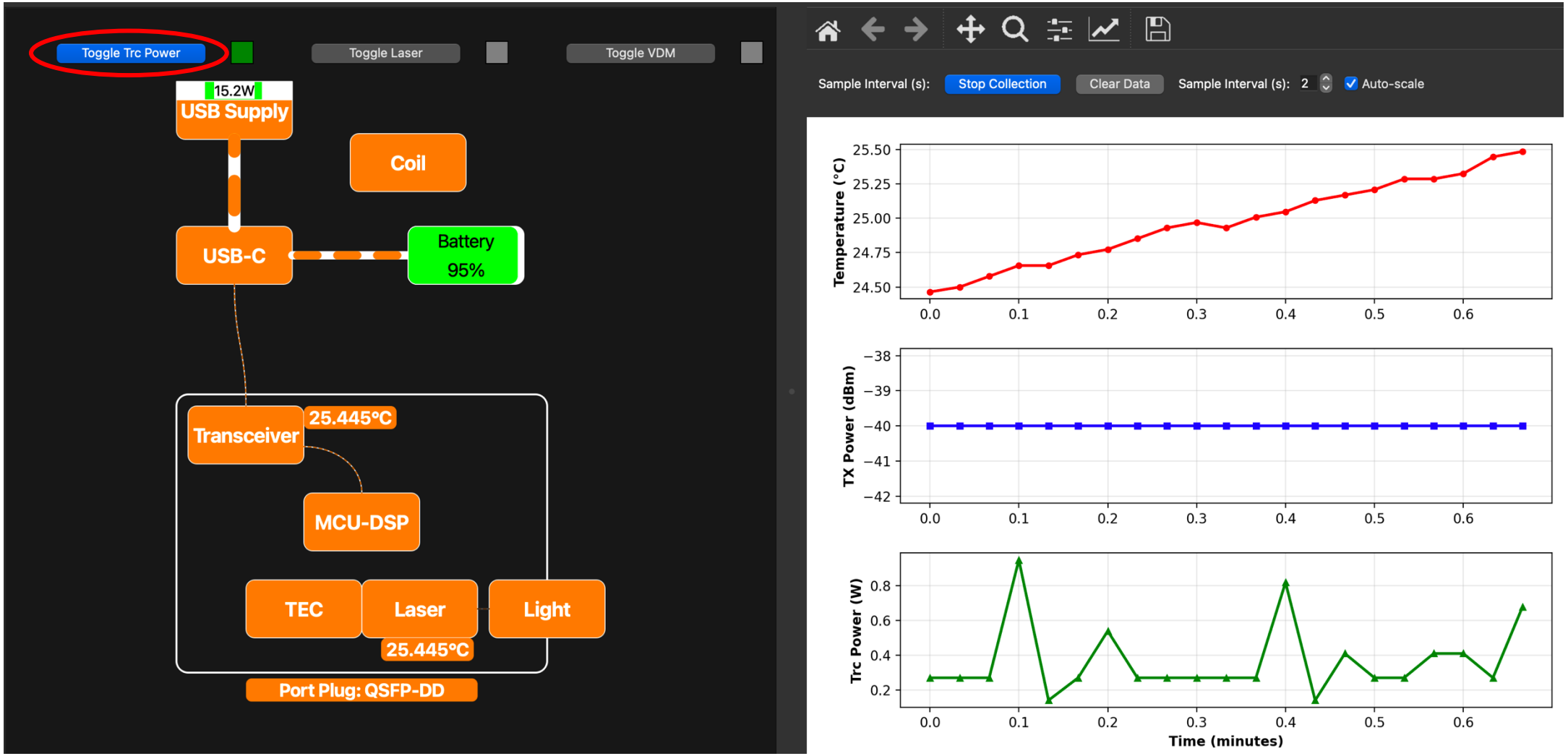


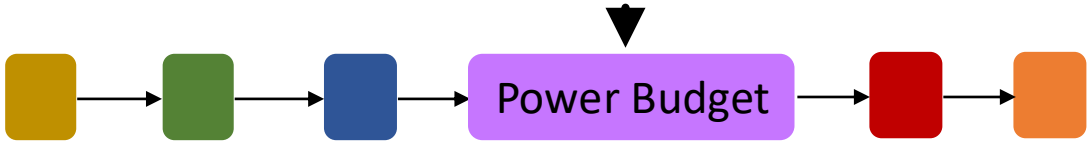
Con Flexbox y herramientos de investigación

*Python, PyQt y Matplotlib en una application

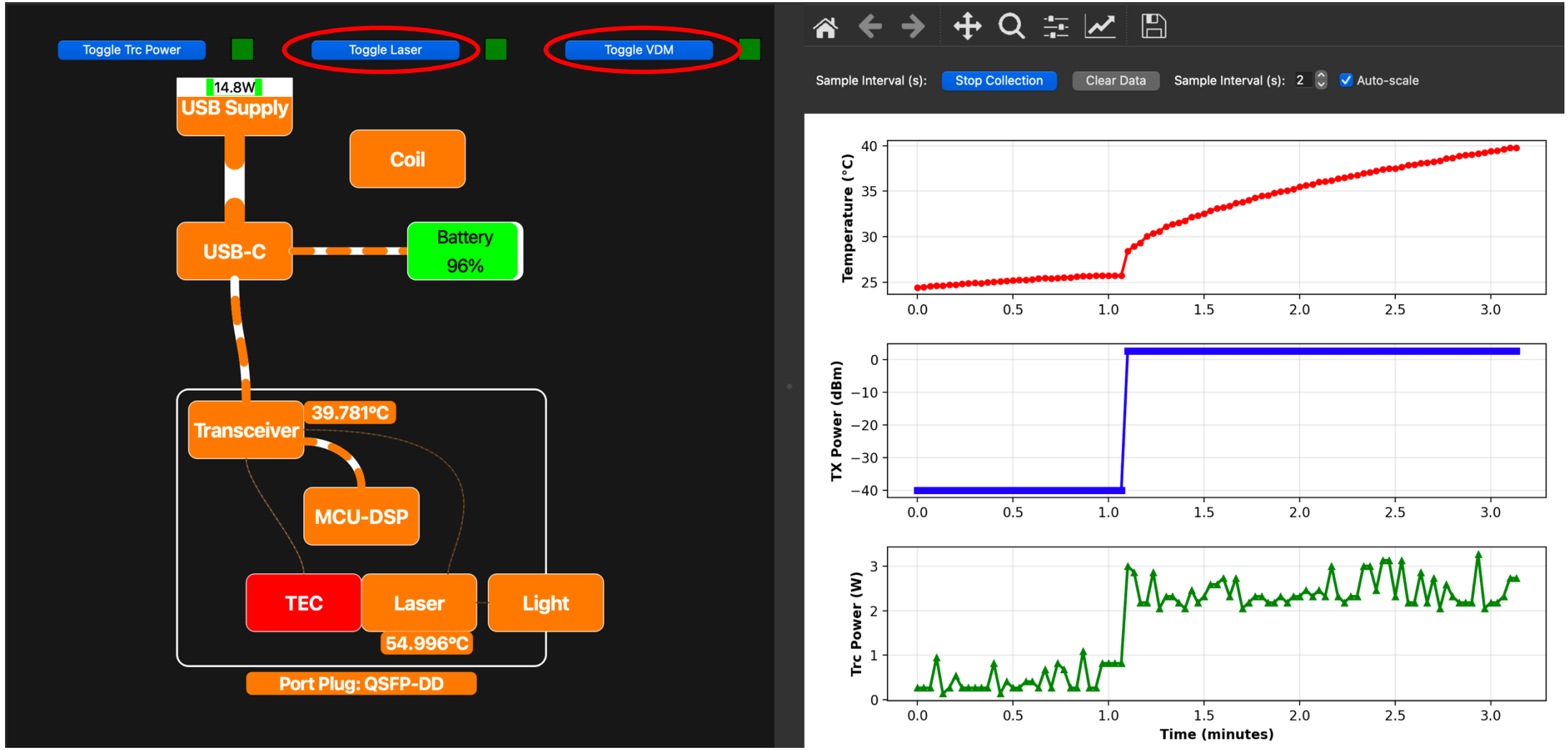


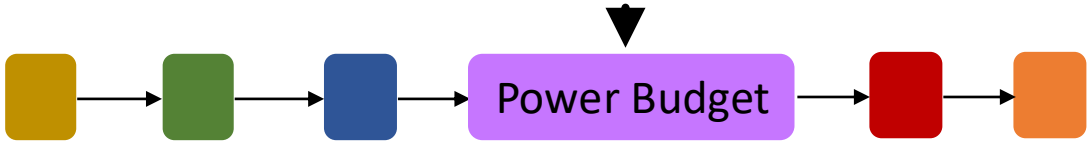
¡Da le energía al transceptor!





Go Go Laser (y VDM) !!

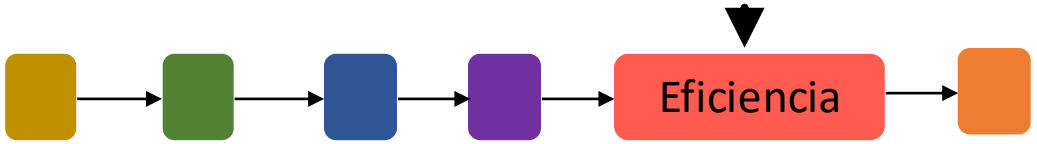




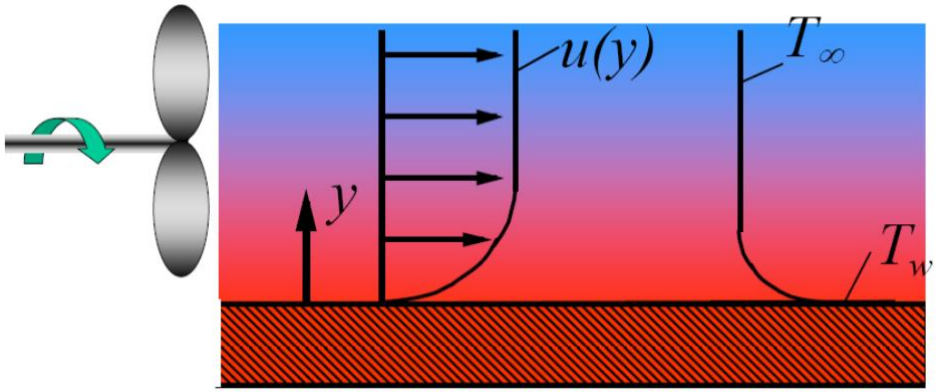
Propuesto energético – Evolución tecnica de los trcs

Consumo (Watts)	2018	2020	2023	2025	2026	Cambios
400G QDD FR4 2km	12	10.5		9		ASIC intercambiado por PAM4 DSP con controlador EML integrado, optimizando de 16 a 7 nm
800G OSFP 2xLR4 Dual Duplex LC			16.5	15.5		Cambio de EML a SiPho
1.6T OSFP224 2xDR4				30	26	Nodo DSP CMOS → Reducción de 5 a 3nm

Source: [17,18]



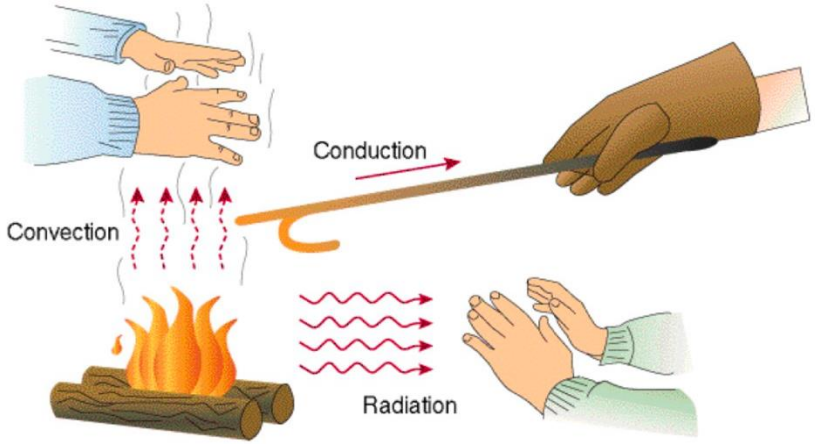
Eficiencia: Como un ventilador transfiere calor

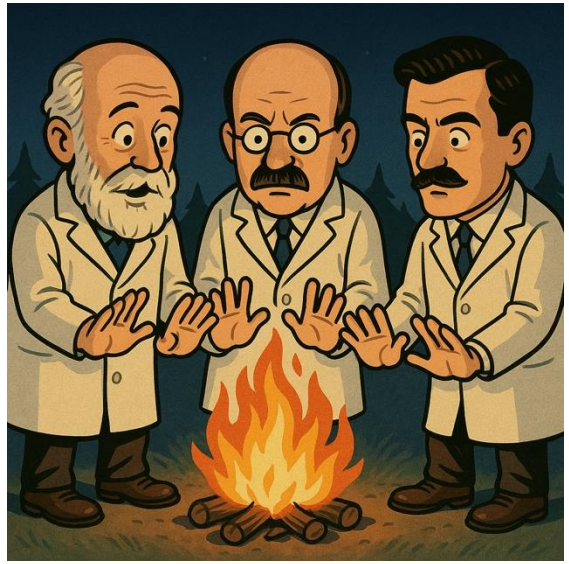


$$P_{Fan} = \frac{\lambda \times Nu}{L} A(T_\infty - T_W)$$

$$Nu = 0.664\sqrt{Re}\sqrt[3]{Pr}$$

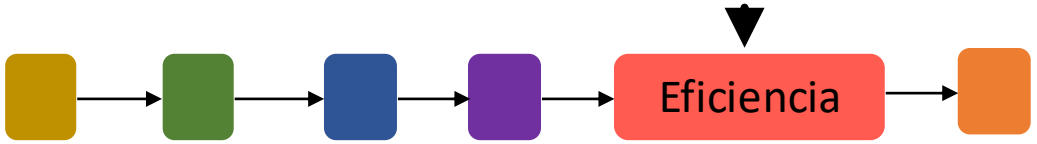
$$Re = \frac{w_\infty L}{\upsilon}$$



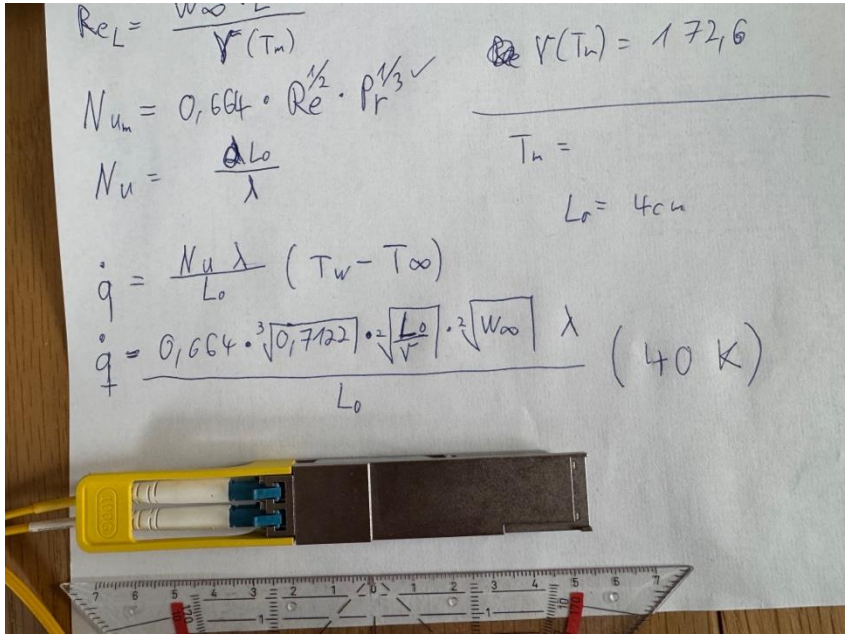


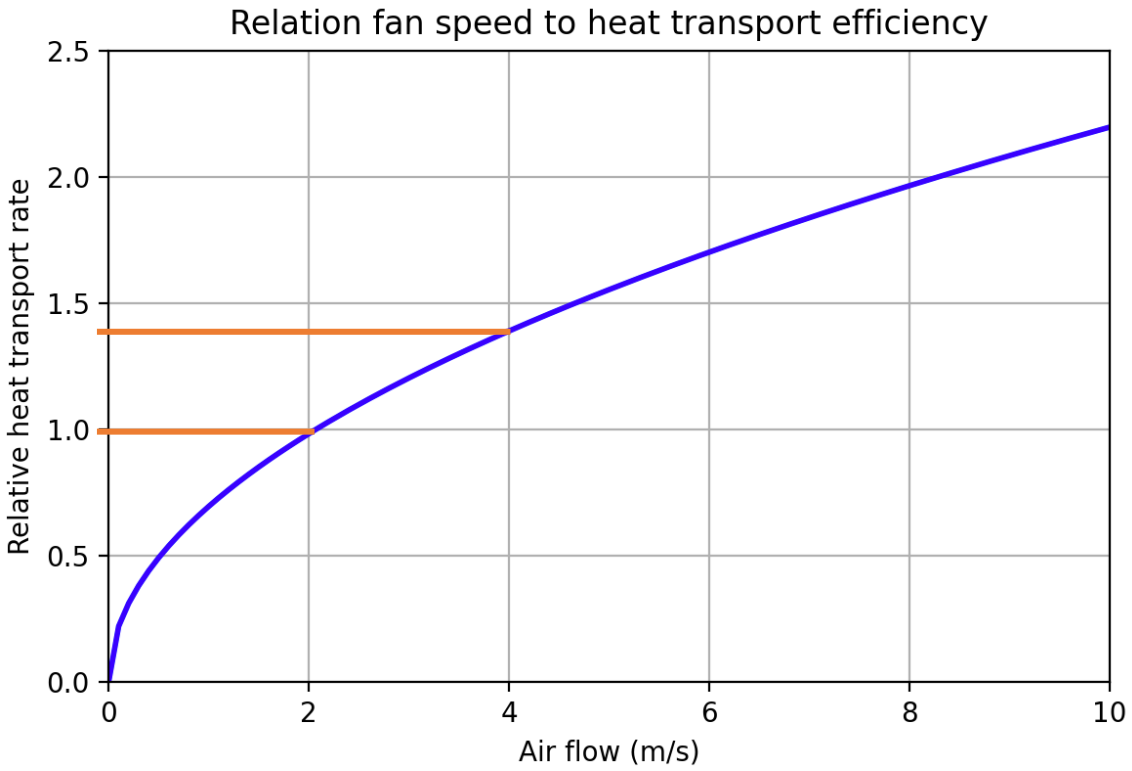
- Nusselt (Nu)
- Prandtl (Pr)
- Reynolds (Re)

Sources: [13]



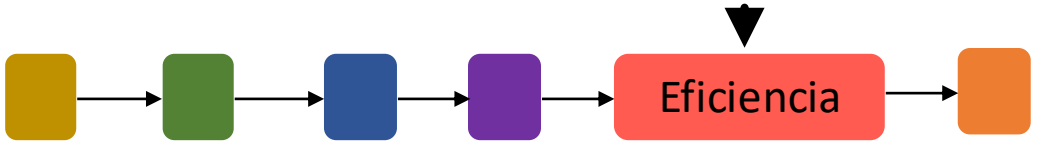
Eficiencia: Como un ventilador transfiere calor





El <u>doble</u> de velocidad
no significa que obtendrás
el <u>doble</u> the Vatios

Sources: [13]

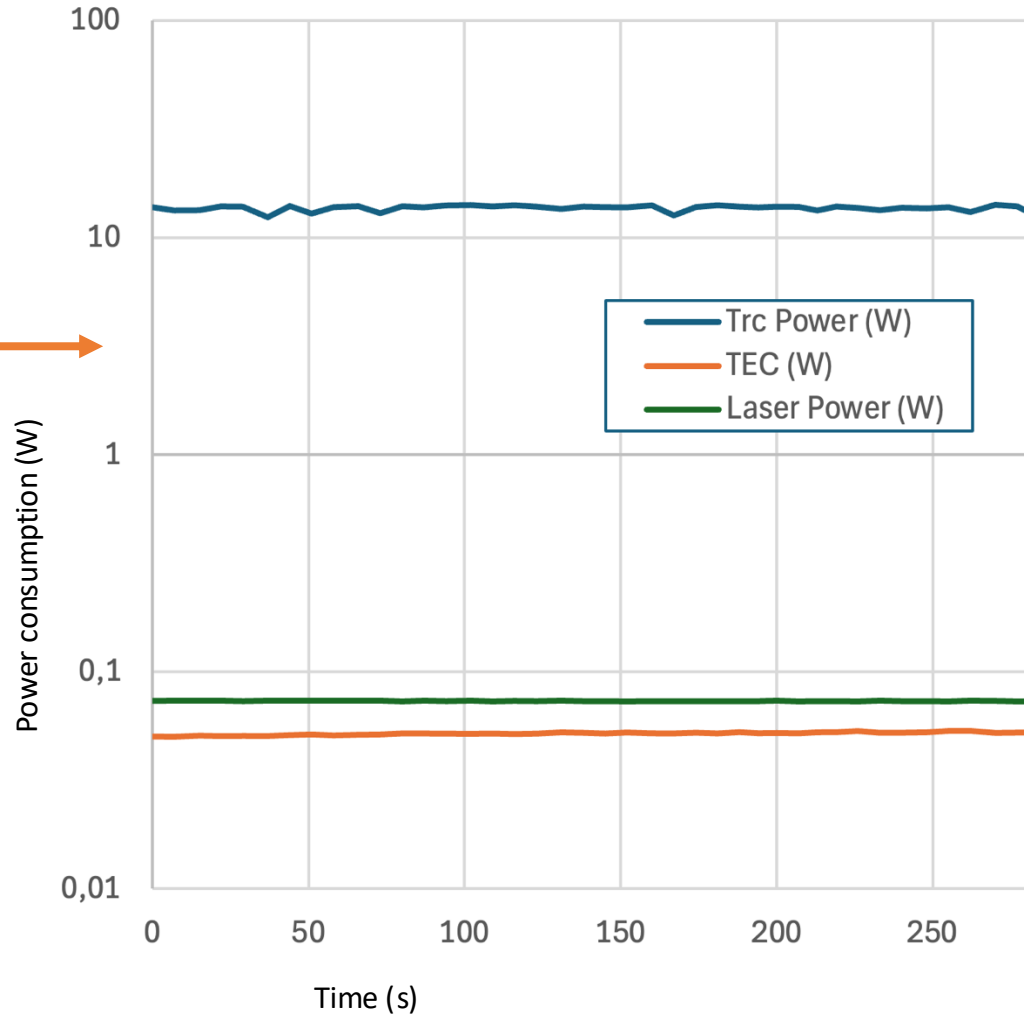


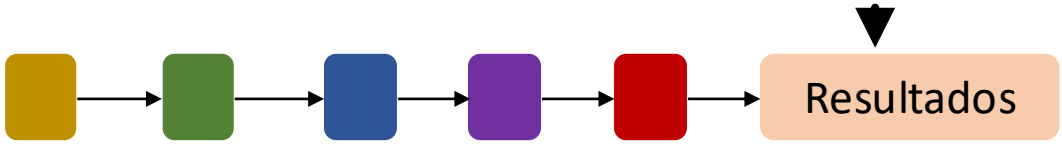
¿Qué influencia tienen TEC y láser?

Power readings de D.CO164HG.2.yT →
400G Coherent con configuración del ventilador a **Q1**

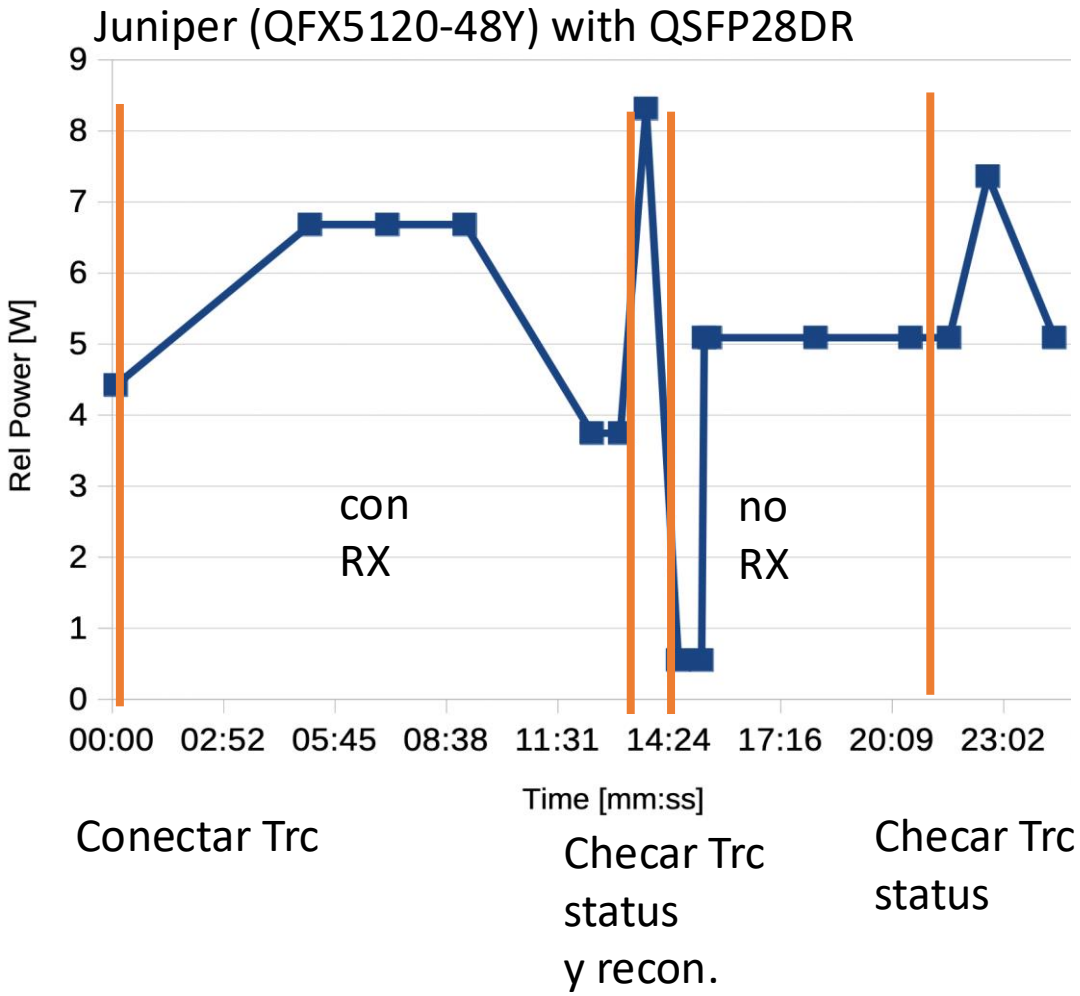
Art No.	Q0 (No Fan)	Q1 (low)	Q2 (mid)	Q3(high)
D.CO164HG.2.yT	75.2	56.49	49.57	47.67

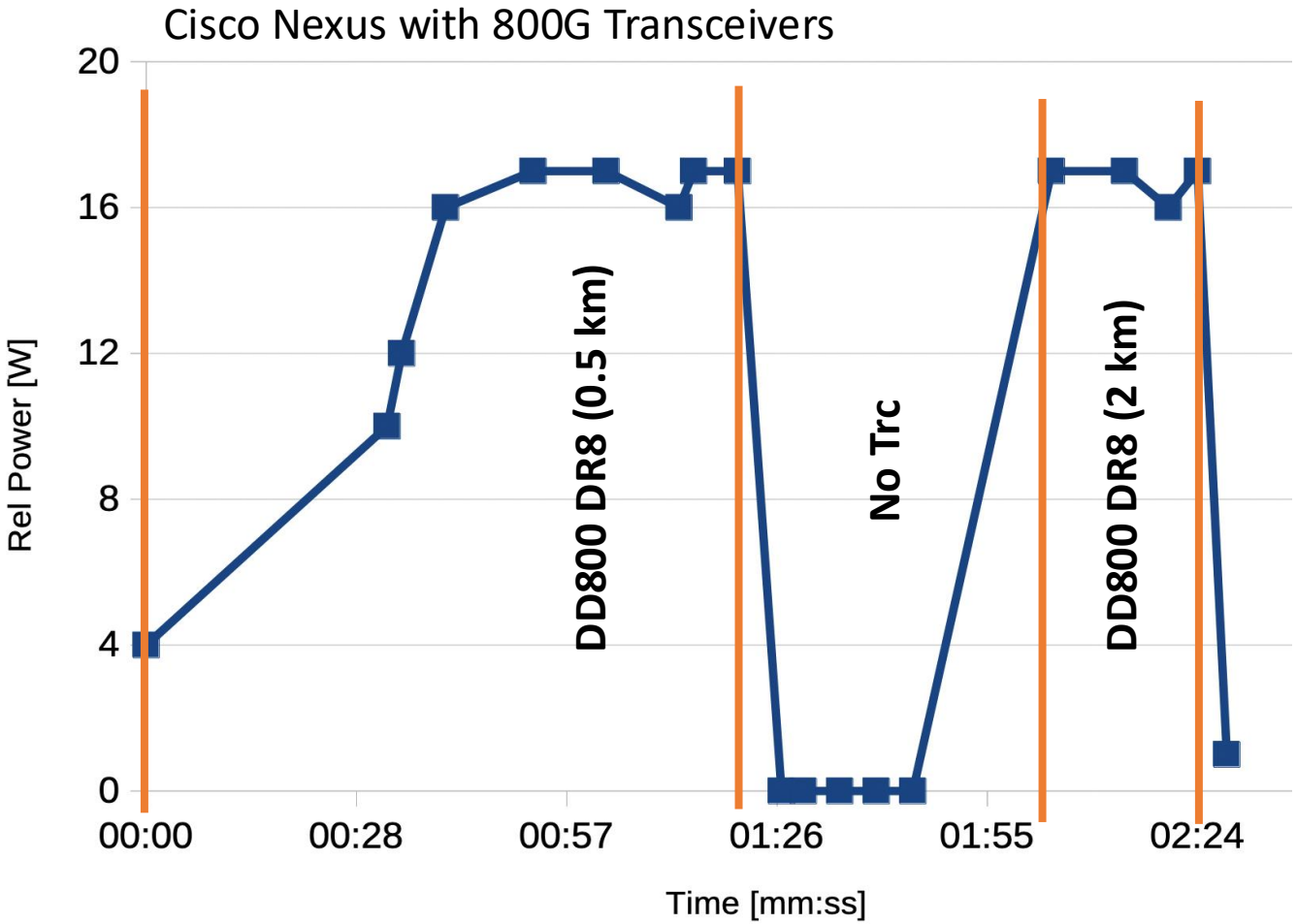
QX	Speed	RPM (approx.)
Q0	0 m/s (obviously)	0
Q1	2.2 m/s	467
Q2	3.0 m/s	637
Q4	3.8 m/s	806





Resultados: Tests con switches y transceptores



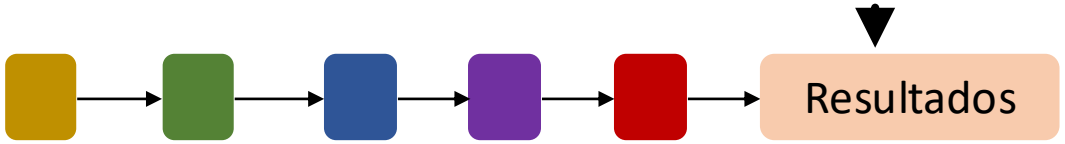


Experimento con un transceptor congelado

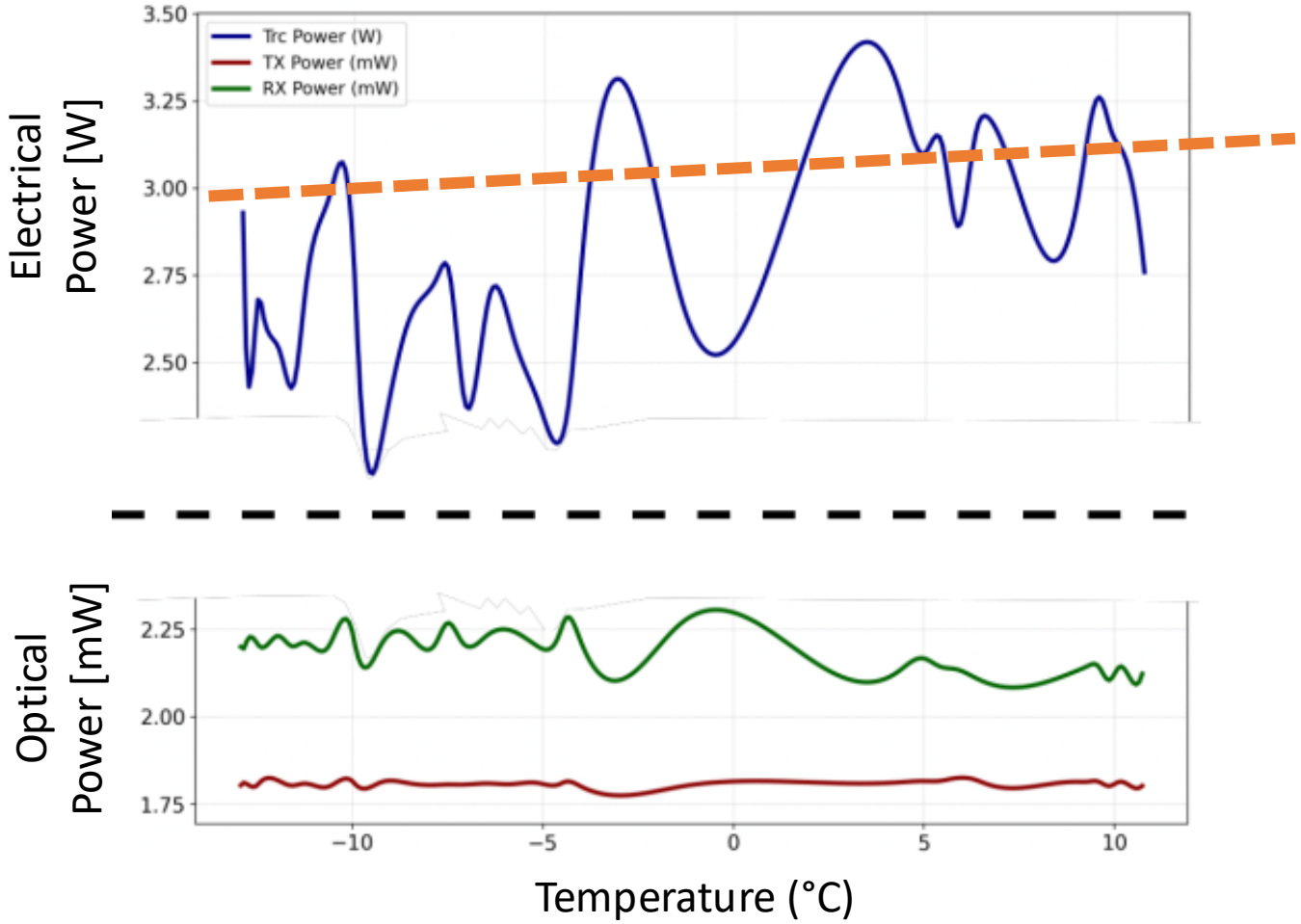
Cubos de hielo

Flexbox 5

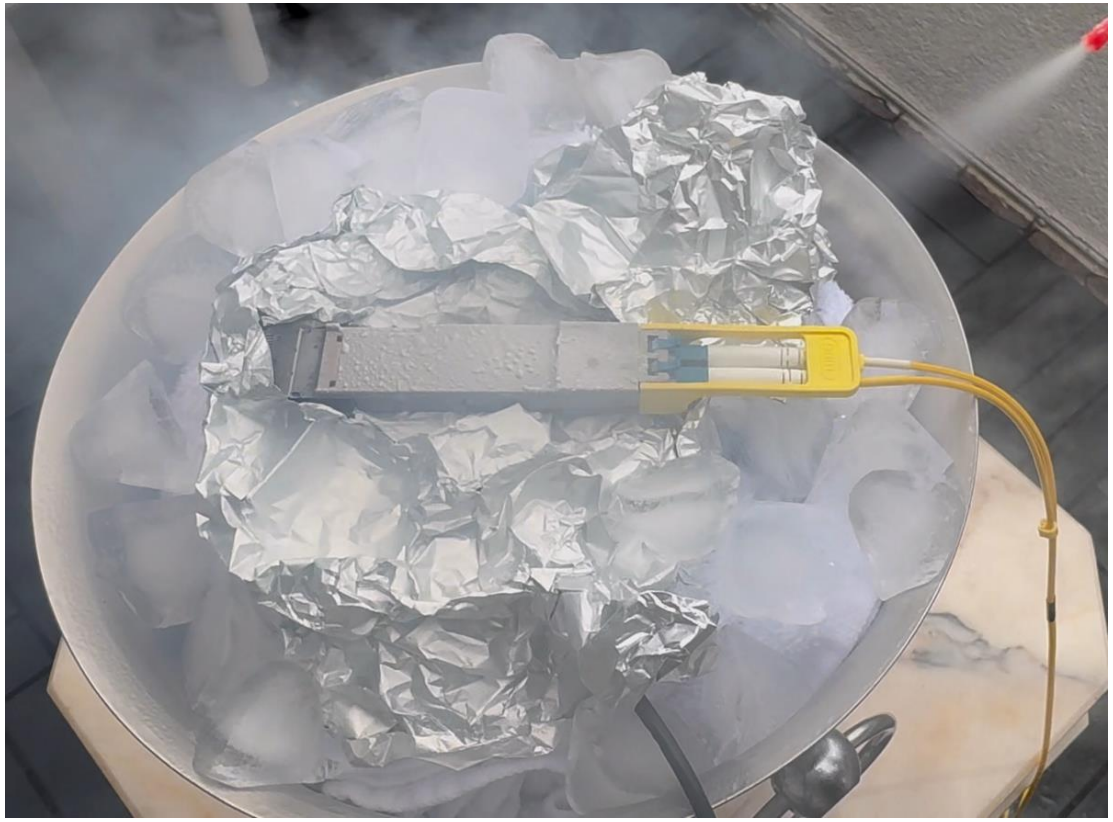
Láminas de aluminio



Resultados: congelado, pero láser calienta desde adentro



Con como 20°C de diferencia tenemos un delta de appróx. 200 mW



Conclusion

- Láser y TEC no son los absorbedores principales de energía. Mejor echar un vistazo a la combinación MCU/DSP y descubrir por medio del PRBS para medir el consumo de energía total.
- Temperatura del láser @ 50 °C → es el punto más eficiente → pero tiene poca influencia al consumo total
  - Sin importar la temperatura del trc, el consumo del láser y TEC se pueden ignorar
- Desactiva tu puerto: El trc será forzado a acceder al modo Low Power y consumirá con un factor entre 7 y 20 menos energía
- Mediendo la energía del switch revelará más detalles
- Puedes medir el consumo de energia poniendo el transceptor en diferentes estados con una Flexbox 5

¡Muchas gracias!

Source: [14]

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